Derivatives: Contracts, Models, and Markets · Technical Note 39

Aggregating Vega in the Standardized Approach

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Section 39.3 of Derivatives: Contracts, Models, and Markets says that vega is aggregated the same way as delta, with a risk weight that grows with the volatility's liquidity horizon, and Example 39.2 applies the rule to a single option. This note sets out what "the same way" means for a book of options: what the risk factors are, how a sensitivity is measured, how two vega risk factors are correlated, and how the three correlation scenarios apply. It then works a two-stock book through every step. The example shows two features of the rule that a single option cannot. Vega on one stock barely diversifies across option maturities, because the prescribed correlation between two maturities is close to one, so a calendar spread is charged on its net vega. And vega on two stocks in the same bucket barely offsets, because the two are correlated only through their delta correlation.

`tools/verify/tn39-vega-aggregation.py` *recomputes every figure here.*

1 RISK FACTORS, SENSITIVITY AND WEIGHT

A vega risk factor is an implied volatility at one of five option maturities: 0.5, 1, 3, 5 and 10 years. For an option on an interest rate the factor has a second dimension, the maturity of the underlying rate or swap, on the same five points, so a swaption book maps onto a grid. Equity factors are grouped into buckets by market capitalization, economy and sector, as for delta.

The sensitivity to factor k is the option's vega multiplied by the implied volatility:

$$s_k = \frac{\partial V}{\partial \sigma_k}\,\sigma_k, \tag{TN39.1}$$

the change in value for a relative change in volatility rather than for a change of one volatility point. Example 39.2's option has a vega of \$28,099.50 per point at a volatility of 25%, so its sensitivity is $28{,}099.50 \times 25 = \$702{,}488$ (a per-point vega times the volatility in points). A position that gains when volatility rises carries a positive sensitivity. Reversing every sign leaves the charge unchanged, so this convention and Example 39.1's loss convention for delta give the same capital.

The weighted sensitivity is $WS_k = RW\, s_k$, with

$$RW = \min\bigl(55\% \times \sqrt{\mathrm{LH}/10},\ 100\%\bigr), \tag{TN39.2}$$

where LH is the liquidity horizon in days of Section 39.4's schedule. The volatility of a large-cap stock has a 20-day horizon and a weight of 77.78%. Every other volatility has a horizon of 40 days or more, and the cap binds from about 33 days, so the weight is 100%.

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2 CORRELATION BETWEEN TWO VEGA RISK FACTORS

Within a bucket, the correlation between factors k and l is the product of two parts:

$$\rho_{kl} = \min\left(\rho_{kl}^{\Delta}\,\rho_{kl}^{\text{opt}},\ 1\right), \qquad \rho_{kl}^{\text{opt}} = e^{-\alpha\,|T_k - T_l|/\min(T_k, T_l)}, \quad \alpha = 1\%. \tag{TN39.3}$$

T_k is the option maturity in years. ρ_{kl}^{Δ} is the delta correlation between the two underlyings: one for two volatilities of the same stock, and the bucket's prescribed figure for two different stocks, which is 25% for two large-cap stocks in an advanced economy. For an interest-rate option the first part, the delta correlation, is replaced by the same exponential in the underlying's maturity, so both dimensions of the grid decorrelate at $\alpha = 1\%$. Buckets are then combined with the delta charge's cross-bucket correlations.

The maturity part is close to one everywhere. At $\alpha = 1\%$ the 6-month and 10-year volatilities of one stock are correlated at $e^{-0.01\times 9.5/0.5} = 0.8270$, and the 1-year and 10-year at 0.9139. Example 39.1's rate curve, at $\alpha = 3\%$, correlates the 2-year and 10-year rates at 0.8869, below the 0.9139 between the 1-year and 10-year volatilities, which lie further apart. The rule treats the volatility term structure as nearly rigid.

The three scenarios are those of the delta charge: the prescribed correlations, $\min(1.25\rho,\ 1)$ and $\max(2\rho - 1,\ 0.75\rho)$, applied to every ρ_{kl}, with the charge set by the largest of the three after the buckets are combined. The bucket charge has the delta form,

$$K_b = \sqrt{\max\left(0,\ \sum_k WS_k^2 + \sum_{k\neq l} \rho_{kl}\, WS_k\, WS_l\right)}. \tag{TN39.4}$$

3 A TWO-STOCK BOOK

Example 3.1 A two-stock vega book in one bucket

A desk holds options on two large-cap stocks, X and Y, in the same advanced-economy bucket. On X it is long volatility at 6 months and 3 years and short at 1 year, a calendar structure with net vega of \$100,000. On Y it is short. The sensitivities, the weighted sensitivities at 77.78%, and the correlations from (TN39.3) are:

Factor	Sensitivity	Weighted
X, 6 months	\$400,000	\$311,127
X, 1 year	−\$600,000	−\$466,690
X, 3 years	\$300,000	\$233,345
Y, 1 year	−\$250,000	−\$194,454

Pair	Prescribed	Low	High
X 6m, X 1y	0.9900	0.9801	1
X 6m, X 3y	0.9512	0.9025	1
X 1y, X 3y	0.9802	0.9604	1
X 6m, Y 1y	0.2475	0.1856	0.3094
X 1y, Y 1y	0.2500	0.1875	0.3125
X 3y, Y 1y	0.2450	0.1838	0.3063

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Between two volatilities of X the low scenario takes $2\rho - 1$; between X and Y it takes 0.75ρ, since $2\rho - 1$ is negative there. Substituting into (TN39.4) gives a bucket charge of \$197,471 under the low correlations, \$192,798 under the prescribed ones and \$188,008 under the high ones. The charge is **\$197,471**, 2.4% above the prescribed case.

Two decompositions show where the charge comes from. Taken alone, stock X has gross weighted vega of \$1,011,163 across its three maturities and net weighted vega of \$77,782, and its charge under the prescribed correlations is \$78,554, only 1.0% above the net. The rule charges the calendar structure almost exactly as if it were a single position of its net vega. A twist in the volatility term structure, the risk that structure actually runs, is nearly free.

Between the stocks the opposite holds. Charged separately, X and Y would need 78,554 + 194,454 = \$273,008; together they need \$192,798, a saving of 29.4%. That is all the offset a 25% correlation allows between X's net long vega and Y's short vega, so most of Y's charge survives. The low scenario binds for the same reason as Example 39.1's butterfly: the book holds offsetting positions, and lowering the correlations weakens the offset.

The weight scales the whole calculation. On small-cap stocks, or on any underlying other than a large-cap stock, the weight would be 100%, every weighted sensitivity would rise by $1/0.7778 - 1 = 28.6\%$, and, with the correlations held fixed, so would the charge, to \$253,878. This is the same 28.6% as Question 7 of Chapter 39 finds for a single option.

4 WHAT THE RULE DOES AND DOES NOT SEE

The vega charge measures a book's exposure to the overall level of each stock's volatility and is nearly blind to its shape across maturities. That matches a rule written to be computed by every bank the same way, but it leaves the term-structure risk of calendar spreads, and of the forward-starting and cliquet structures of Chapter 34 whose value depends on forward volatility, to the internal models approach or to supervisory review. A desk on the standardized approach can run a large volatility term-structure position for very little capital, the kind of exposure Section 39.12's stress tests and supervisory review are left to find.

The cross-stock correlation works the other way. Vega on different stocks in a bucket offsets only through the delta correlation, which is low, so a dispersion book of the kind Chapter 22 describes, long single-stock volatility against short index volatility, is charged nearly in full on each leg: the single-stock legs barely offset one another, and the index volatility sits in a bucket of its own, which offsets them only through a cross-bucket correlation well below one.

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